

Questions

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Q1 - 2024 (04 Apr Shift 1)

If the domain of the function $\sin^{-1}\left(\frac{3x-22}{2x-19}\right) + \log_e\left(\frac{3x^2-8x+5}{x^2-3x-10}\right)$ is $(\alpha, \beta]$, then $3\alpha + 10\beta$ is equal to:

- (1) 100
- (2) 95
- (3) 97
- (4) 98

Q2 - 2024 (04 Apr Shift 1)

Let the sum of the maximum and the minimum values of the function $f(x) = \frac{2x^2-3x+8}{2x^2+3x+8}$ be $\frac{m}{n}$, where $\gcd(m, n) = 1$. Then $m + n$ is equal to :

- (1) 195
- (2) 201
- (3) 217
- (4) 182

Q3 - 2024 (04 Apr Shift 2)

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x}{\sqrt{1+9x^2}}$. If the composition of

$f, \underbrace{(f \circ f \circ f \circ \dots \circ f)}_{10 \text{ times}}(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the value of $\sqrt{3\alpha + 1}$ is equal to _____

Q4 - 2024 (05 Apr Shift 1)

Let $A = \{1, 3, 7, 9, 11\}$ and $B = \{2, 4, 5, 7, 8, 10, 12\}$. Then the total number of one-one maps $f : A \rightarrow B$, such that $f(1) + f(3) = 14$, is :

- (1) 480
- (2) 240
- (3) 120
- (4) 180

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Q5 - 2024 (05 Apr Shift 1)

Let $f(x) = x^5 + 2x^3 + 3x + 1$, $x \in \mathbf{R}$, and $g(x)$ be a function such that $g(f(x)) = x$ for all $x \in \mathbf{R}$. Then

$\frac{g(7)}{g'(7)}$ is equal to :

(1) 14

(2) 42

(3) 7

(4) 1

Q6 - 2024 (05 Apr Shift 1)

The number of distinct real roots of the equation $|x||x+2| - 5|x+1| - 1 = 0$ is _____

Q7 - 2024 (05 Apr Shift 1)

If $S = \{a \in \mathbf{R} : |2a - 1| = 3[a] + 2\{a\}\}$, where $[t]$ denotes the greatest integer less than or equal to t and $\{t\}$ represents the fractional part of t , then $72 \sum_{a \in S} a$ is equal to _____

Q8 - 2024 (05 Apr Shift 2)

Let $f, g : \mathbf{R} \rightarrow \mathbf{R}$ be defined as :

$$f(x) = |x - 1| \text{ and } g(x) = \begin{cases} e^x, & x \geq 0 \\ x + 1, & x \leq 0 \end{cases}$$

Then the function $f(g(x))$ is

(1) neither one-one nor onto.

(2) one-one but not onto.

(3) onto but not one-one.

(4) both one-one and onto.

Q9 - 2024 (06 Apr Shift 1)

The function $f: \mathbf{R} \rightarrow \mathbf{R}$, $f(x) = \frac{x^2 + 2x - 15}{x^2 - 4x + 9}$, $x \in \mathbf{R}$ is

(1) one-one but not onto.

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(2) both one-one and onto.

(3) onto but not one-one.

(4) neither one-one nor onto.

Q10 - 2024 (06 Apr Shift 2)

Let $f(x) = \frac{1}{7 - \sin 5x}$ be a function defined on $\mathbb{R}$. Then the range of the function $f(x)$ is equal to ;

(1) $\left[\frac{1}{7}, \frac{1}{6}\right]$

(2) $\left[\frac{1}{8}, \frac{1}{5}\right]$

(3) $\left[\frac{1}{7}, \frac{1}{5}\right]$

(4) $\left[\frac{1}{8}, \frac{1}{6}\right]$

Q11 - 2024 (08 Apr Shift 1)

Let $[t]$ be the greatest integer less than or equal to t . Let A be the set of all prime factors of 2310 and

$f: A \rightarrow \mathbb{Z}$ be the function $f(x) = \left\lfloor \log_2 \left(x^2 + \left\lceil \frac{x^3}{5} \right\rceil \right) \right\rfloor$. The number of one-to-one functions from A to the

range of f is

(1) 25

(2) 24

(3) 20

(4) 120

Q12 - 2024 (08 Apr Shift 1)

If the range of $f(\theta) = \frac{\sin^4 \theta + 3 \cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$, $\theta \in \mathbb{R}$ is $[\alpha, \beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to _____

Q13 - 2024 (08 Apr Shift 2)

Let $f(x) = \begin{cases} -a & \text{if } -a \leq x \leq 0 \\ x + a & \text{if } 0 < x \leq a \end{cases}$ where $a > 0$ and $g(x) = (f(x) - |f(x)|)/2$.

Then the function $g: [-a, a] \rightarrow [-a, a]$ is

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(1) neither one-one nor onto.

(2) onto.

(3) both one-one and onto.

(4) one-one.

Q14 - 2024 (09 Apr Shift 1)

If the domain of the function $f(x) = \sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is $\mathbf{R} - (\alpha, \beta)$, then $12\alpha\beta$ is equal to :

(1) 32

(2) 40

(3) 24

(4) 36

Q15 - 2024 (09 Apr Shift 1)

If a function f satisfies $f(m+n) = f(m) + f(n)$ for all $m, n \in \mathbf{N}$ and $f(1) = 1$, then the largest natural number λ such that $\sum_{k=1}^{2022} f(\lambda + k) \leq (2022)^2$ is equal to _____

Q16 - 2024 (09 Apr Shift 2)

Let the range of the function $f(x) = \frac{1}{2 + \sin 3x + \cos 3x}$, $x \in \mathbb{R}$ be $[a, b]$. If α and β are respectively the A.M. and the G.M. of a and b , then $\frac{\alpha}{\beta}$ is equal to

(1) π (2) $\sqrt{\pi}$

(3) 2

(4) $\sqrt{2}$

Q17 - 2024 (09 Apr Shift 2)

Let $A = \{(x, y) : 2x + 3y = 23, x, y \in \mathbf{N}\}$ and $B = \{x : (x, y) \in A\}$. Then the number of one-one functions from A to B is equal to _____

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Answer Key

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Solutions

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Q1

$$-1 \leq \frac{3x-22}{2x-19} \leq 1 \quad \frac{3x^2-8x+5}{x^2-3x-10} > 0$$

$$x \in \left(5, \frac{41}{5}\right]$$

$$3\alpha + 10\beta = 97$$

Q2

$$y = \frac{2x^2 - 3x + 8}{2x^2 + 3x + 8}$$

$$x^2(2y-2) + x(3y+3) + 8y-8 = 0$$

use $D \geq 0$

$$(3y+3)^2 - 4(2y-2)(8y-8) \geq 0$$

$$(11y-5)(5y-11) \leq 0$$

$$\Rightarrow y \in \left[\frac{5}{11}, \frac{11}{5}\right]$$

 $y = 1$ is also included

Q3

$$f(f(x)) = \frac{2f(x)}{\sqrt{1+9f^2(x)}} = \frac{4x}{\sqrt{1+9x^2+9 \cdot 2^2x^2}}$$

$$f(f(f(x))) = \frac{2^3x/\sqrt{1+9x^2}}{\sqrt{1+9(1+2^2)\frac{2^2x^2}{1+9x^2}}} = \frac{2^3x}{\sqrt{1+9x^2(1+2^2+2^4)}}$$

 $\therefore$ By observation

$$\alpha = 1 + 2^2 + 2^4 + \dots + 2^{18} = 1 \left(\frac{(2^2)^{10} - 1}{2^2 - 1} \right) = \frac{2^{20} - 1}{3}$$

$$3\alpha + 1 = 2^{20} \rightarrow \sqrt{3\alpha + 1} = 2^{10} = 1024$$

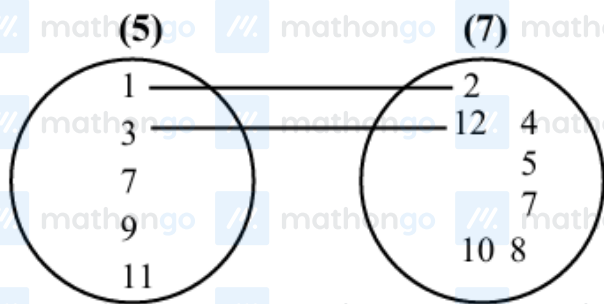
Q4

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Solutions

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$$A = \{1, 3, 7, 9, 11\}$$

$$B = \{2, 4, 5, 7, 8, 10, 12\}$$

$$f(1) + f(3) = 14$$

$$(i) 2 + 12$$

$$(ii) 4 + 10$$

$$2 \times (2 \times 5 \times 4 \times 3) = 240$$

Q5

$$f(x) = x^5 + 2x^3 + 3x + 1$$

$$f'(x) = 5x^4 + 6x^2 + 3$$

$$f'(1) = 5 + 6 + 3 = 14$$

$$g(f(x)) = x$$

$$g'(f(x))f'(x) = 1$$

$$\text{for } f(x) = 7$$

$$\Rightarrow x^5 + 2x^3 + 3x + 1 = 7$$

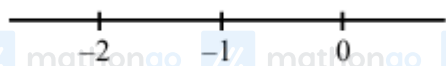
$$\Rightarrow x = 1$$

$$g'(7)f'(1) = 1 \Rightarrow g'(7) = \frac{1}{f'(1)} = \frac{1}{14}$$

$$x = 1, f(x) = 7 \Rightarrow g(7) = 1$$

$$\frac{g(7)}{g'(7)} = \frac{1}{1/14} = 14$$

Q6



Case-1

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Solutions

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$$x \geq 0$$

$$x^2 + 2x - 5x - 5 - 1 = 0$$

$$x^2 - 3x - 6 = 0$$

$$x = \frac{3 \pm \sqrt{9 + 24}}{2} = \frac{3 \pm \sqrt{33}}{2}$$

One positive root

Case-2

$$-1 \leq x < 0$$

$$x^2 - 2x - 5x - 5 - 1 = 0$$

$$x^2 + 7x + 6 = 0$$

$$(x + 6)(x + 1) = 0$$

$$x = -1$$

one root in range

Case-3

$$-2 \leq x < -1$$

$$x^2 - 2x + 5x + 5 - 1 = 0$$

$$x^2 - 3x - 4 = 0$$

$$(x - 4)(x + 1) = 0$$

No root in range

Case-4

$$x < -2$$

$$x^2 + 7x + 4 = 0$$

$$x = \frac{-7 \pm \sqrt{49 - 16}}{2} = \frac{7 \pm \sqrt{33}}{2}$$

one root in range

Total number of distinct roots are 3

Q7

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Solutions

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$$|2a - 1| = 3[a] + 2\{a\}$$

$$|2a - 1| = [a] + 2a$$

$$\text{Case-1 : } a > \frac{1}{2}$$

$$2a - 1 = [a] + 2a$$

$$[a] = -1 \quad \therefore a \in [-1, 0) \text{ Reject}$$

$$\text{Case-2 : } a < \frac{1}{2}$$

$$-2a + 1 = [a] + 2a$$

$$a = I + f$$

$$-2(I + f) + 1 = I + 2I + 2f$$

$$I = 0, f = \frac{1}{4} \quad \therefore a = \frac{1}{4}$$

$$\text{Hence } a = \frac{1}{4}$$

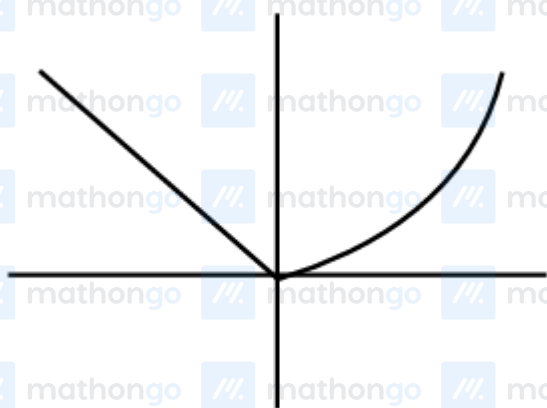
$$72 \sum_{a \in S} a = 72 \times \frac{1}{4} = 18$$

Q8

$$f(g(x)) = |g(x) - 1|$$

$$f \circ g \begin{cases} |e^x - 1| & x \geq 0 \\ |x + 1 - 1| & x \leq 0 \end{cases}$$

$$f \circ g \begin{cases} e^x - 1 & x \geq 0 \\ -x & x \leq 0 \end{cases}$$



Q9

$$f(x) = \frac{(x+5)(x-3)}{x^2 - 4x + 9}$$

$$\text{Let } g(x) = x^2 - 4x + 9$$

$$D < 0$$

$$g(x) > 0 \text{ for } x \in \mathbb{R}$$

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Solutions

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$$\therefore \begin{cases} f(-5) = 0 \\ f(3) = 0 \end{cases}$$

So, $f(x)$ is many-one.

again,

$$yx^2 - 4xy + 9y = x^2 + 2x - 15$$

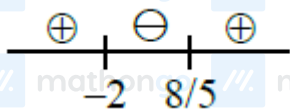
$$x^2(y - 1) - 2x(2y + 1) + (9y + 15) = 0$$

$$\text{for } \forall x \in R \Rightarrow D \geq 0$$

$$D = 4(2y + 1)^2 - 4(y - 1)(9y + 15) \geq 0$$

$$5y^2 + 2y + 16 \leq 0$$

$$(5y - 8)(y + 2) \leq 0$$



$$y \in \left[-2, \frac{8}{5}\right] \text{ range}$$

Q10

$$\sin 5x \in [-1, 1]$$

$$-\sin 5x \in [-1, 1]$$

$$7 - \sin 5x \in [6, 8]$$

$$\frac{1}{7 - \sin 5x} \in \left[\frac{1}{8}, \frac{1}{6}\right]$$

Q11

$$N = 2310 = 231 \times 10$$

$$= 3 \times 11 \times 7 \times 2 \times 5$$

$$A = \{2, 3, 5, 7, 11\}$$

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Solutions

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$$f(x) = \left[\log_2 \left(x^2 + \left[\frac{x^3}{5} \right] \right) \right]$$

$$f(2) = [\log_2(5)] = 2$$

$$f(3) = [\log_2(14)] = 3$$

$$f(5) = [\log_2(25 + 25)] = 5$$

$$f(7) = [\log_2(117)] = 6$$

$$f(11) = [\log_2 387] = 8$$

$$\text{Range of } f : B = \{2, 3, 5, 6, 8\}$$

$$\text{No. of one-one functions} = 5! = 120$$

Q12

$$f(\theta) = \frac{\sin^4 \theta + 3 \cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$$

$$f(\theta) = 1 + \frac{2 \cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$$

$$f(\theta) = \frac{2 \cos^2 \theta}{\cos^4 \theta - \cos^2 \theta + 1} + 1$$

$$f(\theta) = \frac{2}{\cos^2 \theta + \sec^2 \theta - 1} + 1$$

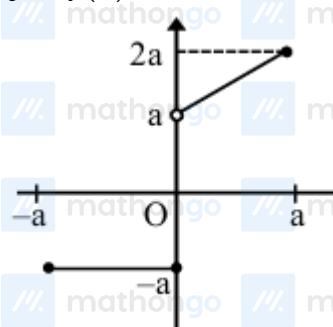
$$f(\theta)|_{\min.} = 1$$

$$f(\theta)|_{\max.} = 3$$

$$S = \frac{64}{1 - 1/3} = 96$$

Q13

$$y = f(x)$$

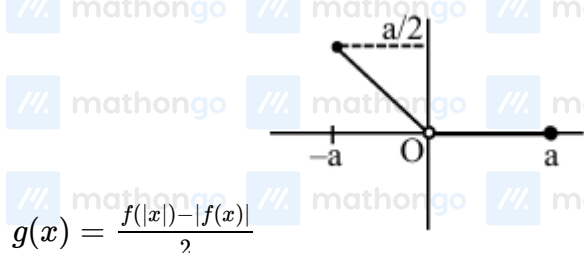
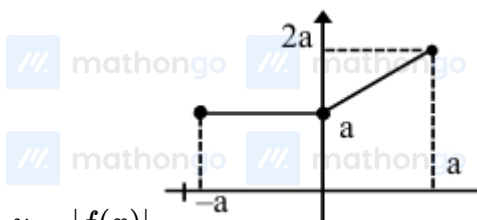
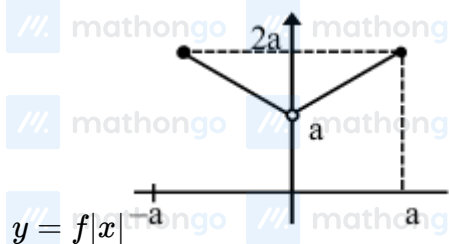


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Solutions

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Q14

Domain of $f(x) = \sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is

$$2x + 3 \neq 0 \text{ and } x \neq \frac{-3}{2} \text{ and } \left| \frac{(x-1)}{2x+3} \right| \leq 1$$

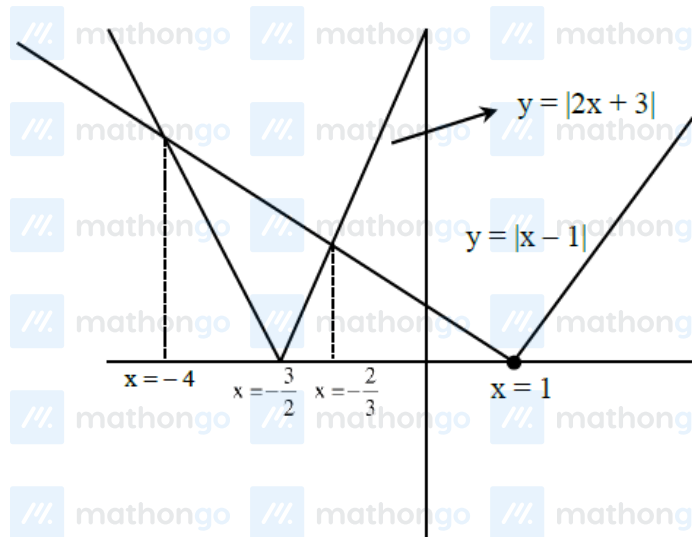
$$|x-1| \leq |2x+3|$$

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Solutions

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$$\begin{aligned} \text{For } |2x + 3| &\geq |x - 1| \\ x &\in (-\infty, -4] \cup \left(-\frac{2}{3}, \infty\right) \\ \alpha &= -4 \text{ and } \beta = -\frac{2}{3} : 12\alpha\beta = 32 \end{aligned}$$

Q15

$$f(m + n) = f(m) + f(n)$$

$$\Rightarrow f(x) = kx$$

$$\Rightarrow f(1) = 1$$

$$\Rightarrow k = 1$$

$$f(x) = x$$

Now

$$\sum_{k=1}^{2022} f(\lambda + k) \leq (2022)^2$$

$$\Rightarrow \sum_{k=1}^{2022} (\lambda + k) \leq (2022)^2$$

$$\Rightarrow 2022\lambda + \frac{2022 \times 2023}{2} \leq (2022)^2$$

$$\Rightarrow \lambda \leq 2022 - \frac{2023}{2}$$

$$\Rightarrow \lambda \leq 1010.5$$

$\therefore$ largest natural no. λ is 1010.

Q16

Solutions

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$$f(x) = \frac{1}{2 + \sin 3x + \cos 3x}$$

$$\left[\frac{1}{2 + \sqrt{2}}, \frac{1}{2 - \sqrt{2}} \right]$$

$$\frac{\alpha}{\beta} = \frac{a+b}{2\sqrt{ab}} = \frac{1}{2} \left(\sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}} \right)$$

$$= \frac{1}{2} \left(\sqrt{\frac{2-\sqrt{2}}{2+\sqrt{2}}} + \sqrt{\frac{2+\sqrt{2}}{2-\sqrt{2}}} \right)$$

$$= \frac{(2-\sqrt{2}) + (2+\sqrt{2})}{2 \times \sqrt{2}} = \sqrt{2}$$

Q17

$$2x + 3y = 23$$

$$x = 1 \quad y = 7$$

$$x = 4 \quad y = 5$$

$$x = 7 \quad y = 3$$

$$x = 10 \quad y = 1$$

$$A \quad B$$

$$(1, 7) \quad 1$$

$$(4, 5) \quad 4$$

$$(7, 3) \quad 7$$

$$(10, 1) \quad 10$$

The number of one-one functions from A to B is equal to 4 !

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